

Junyu (James) Guo

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EDUCATION

- **University of California, Berkeley** Berkeley, CA
Ph.D. in Industrial Engineering and Operations Research *Expected May 2028*
 - **Advisor:** Prof. Javad Lavaei
 - **Selected Coursework:** Advanced Natural Language Processing, Optimization
- **Tsinghua University** Beijing, China
B.S. in Mathematics; GPA: 3.90/4.00; Rank: 3/55 *Sep 2020 – Jun 2024*
 - **Selected Coursework:** Probability Theory, Real Analysis, Complex Analysis, Statistical Inference, Numerical Analysis, and Abstract Algebra
- **Cornell University** Ithaca, NY
Exchange Student; GPA: 4.13/4.30 *Jan 2023 – May 2023*

RESEARCH INTERESTS

- **Machine Learning and Decision-Making:** Reinforcement Learning, Nonconvex Optimization and Large Language Models.
- **Quantitative Methods:** Stochastic Modeling, Monte Carlo Methods, Portfolio Optimization, and Statistical Learning.

RESEARCH EXPERIENCE

- **Optimal Portfolio Allocation in Incomplete Markets** Under Review at Operations Research
Advisor: Prof. Chenxu Li, Peking University, and Prof. Yiwen Shen, HKUST *Sep 2024 – Present*
 - Derived a decomposition of the optimal portfolio policy under CRRA utility and characterized an investor-specific price of risk for incomplete markets.
 - Developed a backward Monte Carlo algorithm to estimate the investor-specific price of risk and compute the resulting portfolio policy, with an extension to high-dimensional settings.
 - Established the algorithm's convergence rate and asymptotic properties using martingale and stochastic-calculus techniques, and validated the results against a closed-form solution under an exponential-volatility model.
- **Diffusion-Regularized Constrained Offline Reinforcement Learning, NeurIPS 2025** Berkeley, CA
Advisor: Prof. Javad Lavaei and Prof. Ming Jin *Oct 2024 – Jun 2025*
 - Developed a constrained offline reinforcement learning method that improves task performance while satisfying safety constraints, without requiring additional interaction with the environment.
 - Modeled the behavior distribution of the offline dataset with a diffusion policy and used its score function to regularize policy optimization, reducing distribution shift toward actions unsupported by the data.
 - Trained reward and safety critics from offline trajectories and designed a gradient-based policy update that balances reward improvement, constraint satisfaction, and closeness to the learned behavior distribution.
 - Evaluated the method across Safety-Gymnasium and Bullet-Safety-Gym tasks, achieving higher returns and lower constraint-violation rates than existing constrained offline reinforcement learning baselines.

INDUSTRY EXPERIENCE

- **TikTok, Data and Privacy Office** San Jose, CA
Machine Learning Engineer Intern *May 2026 – Present*
 - Developed an agentic workflow on TikTok's internal platform to investigate reported production issues by navigating relevant code paths and interacting with the testing environment, producing structured repair plans for engineers.
 - Improved a privacy-review agent for pre-launch features by redesigning its risk-discovery workflow to identify a broader range of potential privacy issues and generate structured reports for the legal team.
- **Sixie Capital** Shanghai, China
Quantitative Research Intern, CTA Strategies *Jun 2023 – Sep 2023*
 - Developed predictive signals from futures price and volume data and evaluated their relationship with subsequent returns through in-sample and out-of-sample backtesting.
 - Defined high-activity "spike trading" regimes and trained LightGBM models to capture nonlinear return-predictive patterns, achieving 6% in-sample and 3% out-of-sample correlation with subsequent returns.
 - Processed high-frequency order-book data on Linux-based research infrastructure to support feature construction, signal evaluation, and backtesting.

SELECTED PUBLICATIONS & PREPRINTS

- **When Context Changes: Understanding Update Failures In LLMs:** 2026. [\[Paper\]](#)
- **LLMs Should Express Uncertainty Explicitly:** 2026. [\[Paper\]](#)
- **Don't Trade Off Safety: Diffusion Regularization for Constrained Offline RL:** NeurIPS 2025. [\[Paper\]](#)
- **StyleBench: Evaluating Thinking Styles in Large Language Models:** 2025. [\[Paper\]](#)
- **Reinforcement Learning for SBM Graphon Games with Re-Sampling:** 2023. [\[Paper\]](#)

HONORS AND AWARDS

- Wolff Fellowship (2024), Berkeley
- First Prize in China Undergraduate Mathematics Competition (2021).
- Scholarship for Merit student of Tsinghua University (2020, 2021, 2022).
- Mathematics Department Academic Scholarship (2021).
- First Prize in China High School National Mathematics Olympic League (2019, 2018).